PN7150 NFC

U1
PN7150B0HN_C11002E-PN7150B0HN_C11002E

+3V3
R1 10K
R2 10K
SDA
SCL

RXP 16
RXN 15
ADR0 1
ADR1 3
SCL 7
SDA 5

VDDD 30
VDD 29
VDDA 28
VDD(TX) 14
VBAT 26
VBAT1 13
VBAT2 12
VDD(TX_IN) 22
VDD(MID) 17
VDD(PAD) 6
C5 1uF
GND
+3V3
C6 1uF
C7 100nF
C8 1uF
C9 100nF
GND

TX1 21
TX2 18
CLK 40
IRQ 8

27.12Mhz
Y1
C3 10pF
C4 10pF
VEN 10
NFC_CLK_XTAL1 36
NFC_CLK_XTAL2 37
GND
GND

VSS(TX) 19
VSS(PAD) 4
VSS
GND

LEDS & BUTTONS

SW1
RESET
RST
GND

GPIO1
GPIO
SW2
GND

R13 1K
D3
LED
LED1
GND

R12 1K
D2
LED
ESP_LED
GND

ANTENA MACHING CIRCUIT

RXP
C23 1nF/16V
L1 160nH
R15 2k2
TX1
100pF/50V
C27
C25 680pF
120pF
C30
R18 3.24R
L3
L
C26 680pF
GND
C31 120pF
R19 3.24R
TX2
L2 160nH
R16 2k2
100pF/50V
C28
RXN
C24 1nF/16V

FLASH

+3V3
10K
R20
QSPI_CS
QSPI_SCK
+3V3
C33 100nF
GND

U6
W25Q16JV
CS
CLK
DI(IO0)
DO(IO1)
IO2
IO3
VCC
GND
EP
GND

QSPI_DATA0
QSPI_DATA1
QSPI_DATA2
QSPI_DATA3

MICROCONTROLLER

U2
+1V2
+3V3
DVDD DVDD VREG_VOUT VREG_IN IOVDD IOVDD IOVDD IOVDD IOVDD IOVDD USB_VDD ADC_AVDD

USB_DP
USB_DM
RD+
RD-
R9 27
R10 27
D+
D-

QSPI_CS 56
QSPI_DATA0 53
QSPI_DATA1 55
QSPI_DATA2 54
QSPI_DATA3 51
QSPI_SCK 52
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK

GPIO0 2
GPIO1 3
GPIO2 4
GPIO3 5 ESP_RESET
GPIO4 6
GPIO5 7
GPIO6 8 VEN
GPIO7 9 IRQ
GPIO8 11 CIPO_TX
GPIO9 12 CS_RX
GPIO10 13 ACK_CTS
GPIO11 14 COPI_RTS
GPIO12 15 SDA
GPIO13 16 SCL
GPIO14 17 SCK
GPIO15 18
GPIO16 27
GPIO17 28 GPIO1
GPIO18 29 IN_B
GPIO19 30 IN_A
GPIO20 31 ESP_GPIO0
GPIO21 32
GPIO22 34
GPIO23 35
GPIO24 36
GPIO25 37
GPIO26_ADC0 38 B_ADC
GPIO27_ADC1 39 LED1
GPIO28_ADC2 40
GPIO29_ADC3 41

20 XIN
21 XOUT
1K R3
Y2 12Mhz
C1 27pF
C2 27pF
GND
GND
RST 26 RUN
SWCLK 24 SWCLK
SWDIO 25 SWD
TESTEN 19
GND 57
GND
GND
RP2040

Raspberry Pi
RP2040

JP1
BOOT
GND
QSPI_CS

+3V3 C10 1uF GND
+3V3 C12 100nF GND
+3V3 C13 100nF GND
+3V3 C15 100nF GND
+3V3 C17 100nF GND
+3V3 C19 100nF GND
+3V3 C20 100nF GND
+3V3 C21 100nF GND

+1V2 C14 1uF GND
+1V2 C16 100nF GND
+1V2 C18 100nF GND

WIFI & BLE NINA

+3V3
R17 10K
ESP_RESET
+3V3
U5
ESP32-WROOM-32D
EN
VDD
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
GND

IO0 25 ESP_GPIO0
TXD0/IO1 35 CS_RX
IO2 24
RXD0/IO3 34 CIPO_TX
IO4 26
IO5 29 CS_RX
IO12 14 COPI_RTS
IO13 16
IO14 13
IO15 23
IO16 27
IO17 28
IO18 30 SCK
IO19 31
IO21 33
IO22 36
IO23 37 CIPO_TX
IO25 10 ESP_LED
IO26 11
IO27 12
IO32 8
IO33 9 ACK_CTS
IO34 6
IO35 7

+3V3
C32 100nF
GND

COIL

VBUS
+BATT
SW4A
SOURCE MAG

U7
TC4424
NC NC
IN_A IN_A OUT_A OUTA
GND VDD
IN_B IN_B OUT_B OUTB
GND
C34 10uf
GND

OUTA J3
OUTB MAGSPOOF

REGULATOR AND BATTERY

P1
USB_C_Plug_USB2.0
VBUS A4
CC A5 R4 5k1
VCONN B5 R5 5k1
D- A7 D-
D+ A6 D+
SHIELD
GND
S1
A1
GND

F1 0.5A
VBUS

+BATT
R6 100K
B_ADC
R7 100K
GND

VBUS
U3
MCP73831-3-OT
VDD
VSS
PROG
VBAT
STAT
+BATT
C11 10uF
R8 5.1K
R11 1K GND
GND
GND
VBUS
D1
LED

+BATT
J1
Battery
GND

VBUS
+BATT
Q1
DMG3415U-7
D4
MBR120
R14 100K
GND
ON
SW3
C22 10uF
GND
U4
AP2112K-3.3
VIN
EN
VOUT
GND
GND
+3V3
C29 10uF
GND

ADDRESS SELECT

R21 100K ADR0
GND
R22 100K ADR1
GND

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

Andres Sabas
Electronic Cats
Sheet: /
File: working.kicad_sch

Title: Bomber Cat

Size: A3	Date: 2023-01-06	**Rev: 1.2**
KiCad E.D.A. eeschema 7.0.6		Id: 1/1